2. [Maximum mark: 6]

(a) Given that $x > 7$, show that $\frac{x}{x^2 - 8x + 7} \times \frac{x^2 - 1}{x + 1} \equiv \frac{x}{x - 7}$. [2]

(b) Hence, or otherwise, solve $\log_2[x(x^2 - 1)] - 1 = \log_2[(x^2 - 8x + 7)(x + 1)]$. [4]